

Template Test

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I

Test

Part I: Contents

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1 A test chapter

§1.1 Lines

Prototypical example for this section: Fit $y = a + bx$ to five points.

A **least-squares line** minimises squared residuals.

Theorem 1.1.1 (Slope)

$$\hat{b} = \sum (x_i - \bar{x})(y_i - \bar{y}) / \sum (x_i - \bar{x})^2.$$

Example 1.1.2 (Five points)

Slope 0.60, intercept 2.20.

code/test/t01_demo.py

```
1 x = np.array([1, 2, 3, 4, 5], dtype=float)
2 y = np.array([2, 4, 5, 4, 5], dtype=float)
3 xb, yb = x.mean(), y.mean()
4 slope = ((x - xb) * (y - yb)).sum() / ((x - xb) ** 2).sum()
5 intercept = yb - slope * xb
```

output (generated by build.sh)

slope=0.60 intercept=2.20

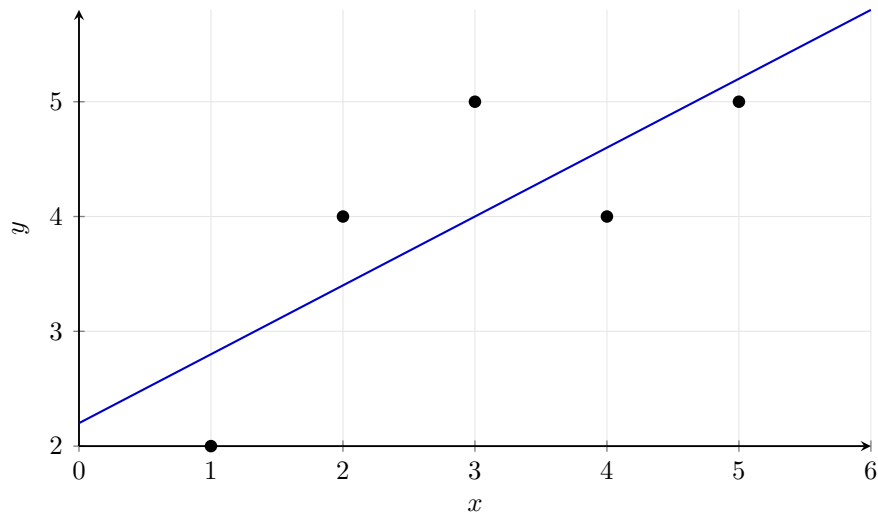
Remark 1.1.3 — Trap. Slope is not $\bar{y}/\bar{x}$.

Question 1.1.4. Why squared?

Least squares is a projection.

Algorithm: OLS

1. centre
2. divide



Stub check: [STUB: remove me]. Missing: **??[nope/x]**.

§1.2 A few problems to think about

Problem 1A (Easy). Compute the slope.

hint · solution



Problem 1B. Prove it.

hint · solution



A Hints

1A. Centre first.

1B. Differentiate.

B Solutions

1A. It is 0.60.

1B. Set derivative to zero.